

Natalia Espinosa Dice

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Education

Expected May 2026 **B.S.E, Computer Science, Princeton University**
Advisor: Prof. Tom Griffiths
Major in Computer Science with Minor in Cognitive Science (GPA: 3.9)

Research Experience

- 2025–Present **Computational Cognitive Science Lab, Princeton University**
Advisor: Prof. Tom Griffiths
Research Areas: Reinforcement Learning
- Developed curriculum-based reinforcement learning methods and implemented progressive neural networks to mitigate catastrophic forgetting across stages.
 - Leading an ongoing project on hierarchical reinforcement learning, in which modular developmental skills are trained individually and coordinated by a high-level value-based controller, to enable scalable and adaptive behavior in complex environments.
 - Investigating how human motor development can provide useful inductive biases for skill composition in humanoid locomotion, where naive exploration is infeasible.
- Summer 2025 **Health Information Privacy Lab, Vanderbilt University**
Advisor: Prof. Brad Malin
Research Areas: Synthetic Data Generation, Reinforcement Learning
- Led a project to design an inverse reinforcement learning-based generative model for privacy-preserving synthetic health data under limited training samples.
 - Benchmarked and achieved competitive performance against existing state-of-the-art generative adversarial networks and diffusion models.
- Fall 2024 **Independent Work, Princeton University**
Advisor: Prof. Xiaoyan Li
Research Areas: Machine Learning for Social Science
- Built XGBoost and Random Forest models to forecast Colombian drug trafficking hotspots using socioeconomic indicators and engineered time-series features.
 - Identified key drivers such as crime rates and urban-rural population dynamics, offering an empirical analysis of existing sociopolitical theories.
- Summer 2024 **School of Information Sciences, University of Macedonia**
Advisors: Prof. Eftichios Protopapadakis, Prof. Christine Syriopoulou-Delli
Research Areas: Deep Learning, Representation Learning
- Led a project developing stacked autoencoders to learn latent representations for distinguishing autism from typical development within structural MRI data.
 - Investigated group-level neural differences by analyzing reconstruction error patterns across cross-tested models.

Industry Experience

- 2023–2024 (Part Time) **Research Intern, Dasion**
Advisor: Dr. Weiqing Gu
- Developed and refined a machine learning model for voice-based medical diagnosis that was adopted into ongoing company workflows.
 - Extended models to diseases including depression and diabetes.
- Summer 2023 **Machine Learning Engineer Intern, Dasion**
- Developed machine learning pipelines for voice-based medical diagnosis of autism, including robust preprocessing and advanced audio feature extraction.

Publications

- [1] N.J. Jackson, **N. Espinosa Dice**, C. Yan, and B. Malin. *Reward-Guided Generation Improves the Scientific Utility of Synthetic Biomedical Data*. Under review, 2026.
- [2] **N. Espinosa Dice**, N.J. Jackson, C. Yan, A. Lee, and B. Malin. *A Reinforcement Learning Approach to Synthetic Data Generation*. Under review, 2025.

Teaching and Outreach

- 2024–
Present **Head Fellow**, *Princeton University*, Writing Center
 - Hold one-on-one writing conferences with undergraduate and graduate students across disciplines.
 - Mentor and supervise a cohort of 7 fellows, and lead training for incoming fellows.
 - Lead workshops on scientific writing and host events introducing first-years to the research and writing process.
- 2024–
Present **Editor-In-Chief**, *Princeton University*, Tortoise Journal
 - Direct the publication of the Writing Center’s annual pedagogy journal, a fully student-run initiative.
 - Contribute editorial commentaries published alongside student work.
- Spring
2026 **Head Course Assistant**, *Princeton University*, SML 354 Machine Learning
 - Head undergraduate course assistant for SML 354: Machine Learning: A Practical Introduction.
 - Grade assignments, hold weekly office hours and assist with course logistics.
- Fall
2025 **Mentor**, *Princeton University*, Engineering Council Mentorship Program
 - Advise underclass engineering students on course planning and career development.
- Spring
2025 **Course Assistant**, *Princeton University*, Intro. to Machine Learning
 - Graded assignments and held weekly office hours to support student learning.
- Fall
2024 **Course Assistant**, *Princeton University*, Data Structures and Algorithms
 - Graded assignments and provided feedback to support student learning.
- 2023–2024 **Fellow**, *Princeton University*, Writing Center
 - Conducted 50+ one-on-one writing conferences per semester with undergraduate and graduate students.
 - Completed extensive mentorship training and professional development workshops.
- 2023–2024 **Editor**, *Princeton University*, Tortoise Journal
 - Selected and edited submissions in collaboration with student authors.
- 2022–2023 **College Mentor**, *UStrive*
 - Mentored two high school students through the college application process.
 - Provided guidance on admissions strategy and detailed feedback on essays.

Honors

- 2025 **Tau Beta Pi**, *National Engineering Honor Society*, Princeton University
- 2025 **Selected Student Panelist**, *Ivy-Plus Writing Consortium Conference*